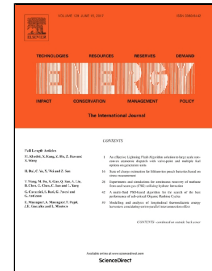


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Research Highlights

- **A novel optimal placement strategy of electric vehicles charging station.**
- **A hybrid algorithm based on both GA and PSO.**
- **Voltage profile shown improvement in the lowest p.u. voltage value.**
- **A better performance in terms of quality of solution with lesser number of iterations.**
- **The minimum stress on the distribution system.**

Optimal Planning of Electric Vehicle Charging Station at the Distribution System Using Hybrid Optimization Algorithm

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Abstract: India's ever increasing population has made it necessary to develop alternative modes of transportation with electric vehicles being the most preferred option. The major obstacle is the deteriorating impact on the utility distribution system brought about by improper setup of these charging stations. This paper deals with the optimal planning (siting and sizing) of charging station infrastructure in the city of Allahabad, India. This city is one of the upcoming smart cities, where electric vehicle transportation pilot project is going on under Government of India initiative. In this context, a hybrid algorithm based on genetic algorithm and improved version of conventional particle swarm optimization is utilized for finding optimal placement of charging station in the Allahabad distribution system. The particle swarm optimization algorithm re-optimizes the received sub-optimal solution (site and the size of the station) which leads to an improvement in the algorithm functionality and enhances quality of solution. The genetic algorithm and improved version of conventional particle swarm optimization algorithm will also be compared with a conventional genetic algorithm and particle swarm optimization. Through simulation studies on a real time system of Allahabad city, the superior performance of the aforementioned technique with respect to genetic algorithm and particle swarm optimization in terms of improvement in voltage profile and quality.

Index terms: Charging Stations, Electric Vehicle (EV's), Genetic Algorithm (GA), Particle Swarm Optimization (PSO).

NOMENCLATURE

IC_i Initial development cost of i^{th} charging station

C_p	Rated power of connector
C_{dev}	Development cost of connector
C_{land}	Rental cost for 5 years
N_i	Number of connectors in i^{th} charging station
S_{cap}^i	Total power capacity of i^{th} charging station
$PLRI_i$	Active Power Loss Reduction Index of i^{th} charging station
$P_{L(base)}$	Active power loss before connection of charging station
$P_{L(CSi)}$	Active power loss after connection of i^{th} charging station
$QLRI_i$	Reactive Power Loss Reduction Index of i^{th} charging station
$Q_{L(base)}$	Reactive power loss before connection of charging station
$Q_{L(CSi)}$	Reactive power loss after connection of i^{th} charging station
$VPII$	Voltage Profile Improvement Index
α	Scaling value
X	Binary variable(0 or 1)
NC	Prospective charging stations to be setup

1. INTRODUCTION

Electric vehicle (EV) and Plug-in Hybrid Electric Vehicle (PHEV) market in India has expanded radically over the course of the last few years. Environmental awareness against increasing greenhouse gas emissions [1], financial savings in term of fuel consumption, and incentives provided by the Indian Government are the major contributors for the shift in mindset of the public. Government of India propelled the National Electric Mobility Mission Plan (NEMMP) 2020 in 2013 which exemplified strategies, objectives and incentives for the growth in sales of hybrid and electric vehicles with a set focus of 7 million from year 2020 onwards [2].

While battery charger technologies are crucial in the development of EV industry, the deployment of optimally sized and located charging stations is the first challenge which is to be encountered primarily. Charging power levels are divided into three levels. Level 1 and Level 2 constitute the low and semi fast charging up to power level of 1.9 kW and 19 kW respectively. Level 3 charging is the fastest charging method with charging power levels reaching up to 100 kW [3]. Such charging stations can be situated either

at exclusive private based (level-I) or commercially owned public parking lots, garages, working place etc. (level II or III). The major obstacle is the deteriorating impact on the utility distribution system brought about by improper setup of these charging stations [4-6]. The problems due to integration of electric vehicles in a renewable energy based microgrid were discussed in [4]. The scenario of electric taxis was discussed in the capital city of China, Beijing in [5]. In [6], a global perspective was given on the energy crisis arising due to electric vehicles. The problem of optimal placement of electric vehicle charging stations (EVCS) has been dealt with extensively in academia over the last few years [7-110]. In [7], the issue of sizing and location of EVCS was taken up separately, whereas in [8], a combined optimization, joint approach was followed by applying Particle Swarm Optimization (PSO) to minimize the objective function. Initial screening of the candidate sites was done based on environmental factors (such as reliable power supply, land cost, load locations etc.) and reachable radius of EVCS's. Another interesting methodology followed in [9] utilized the concepts of graph theory to formulate the EVCS placement objective function and provided four different methods to solve it. The problem was deemed NP (Non-Deterministic Polynomial Time)-hard. A comparative analysis investigating the pros and cons of each method was performed. Another study was taken up in Ontario region of Canada, which analyzed the zonal model of Ontario's transmission network for optimal placement of EV's and PHEV's, keeps the penetration level within an acceptable limit [10]. The concept of battery charge/swap stations was also produced in [11]. An enterprise by the name of 'Better Place' based in California provides battery charge/swap services. The problem of handling such heavy battery packs (minimum 300 kg), additional liability and the huge capital expense makes it difficult to implement on a large scale [12].

Penetration levels of EVs has been a matter of concern for researchers and industry professionals. Higher penetration levels pose various challenges at local distribution and grid level [13]. Renewable based energy systems with built-in EV chargers would be a normal feature of the future smart grid. In order to ensure that existing loads are supplied power optimally while ensuring charging of EVs, a stochastic algorithm was devised in [14]. Another major issue that crops up with setting up of an EV charging infrastructure is the carbon emissions level. Scheduling charging routines for EVs and V2G infrastructure was dealt in [15].

In [16], a three objective problem has been solved by data envelopment method enhanced by cross entropy method for increasing speed and efficiency. The objective function also minimizes voltage deviation and power loss in the distribution system. Similar research has been carried out in districts of Iran [17]. In [18], Cplex software was used to solve the siting and sizing problem in Stockholm, Sweden. Another study conducted in Germany [19], used OLoCs (Optimized Location Scheme for electric charging stations) which was useful in reducing the convergence time. The hybrid algorithm combined GA

and PSO to solve the problem of optimal siting of DG's in a distribution system [21]. The most popular form of public transportation in the city is e-rickshaw.

The present work proposes a novel CS deployment technique and using the same framework as in [21], the Genetic Algorithm based Improved Particle Swarm Optimization (GAIPSO) has been used in this paper to tackle the problem of optimal siting and sizing of electric vehicle charging stations in Allahabad city, India. The initial population for establishing the charging station is produced. For the given objective function, GA is used to produce sub-optimal size and site for each chosen CS which is then passed on to the PSO, hence the name IPSO. IPSO particles are initialized which re-optimize the obtained solution. Hence the dual optimization produces better quality results and in lesser iteration as compared to GA and PSO. To the best of author's knowledge, the proposed work is the first of its kind for any city in India. Section 2 encapsulates the mathematical approach to formulating the objective function specific to the city of Allahabad while giving due consideration to consumer satisfaction and utility power importing limitations. Section 3 deals with the proposed algorithm and its implementation issues. Section 4 includes the case study on a real time bus system in city of Allahabad.

2. OBJECTIVE FUNCTION AND PROBLEM FORMULATION

The problem of placing the charging station in the city is multidimensional in nature. The objective function considers the economic aspect of station setup such as land cost, station equipment, operating and maintenance cost. The cost of equipment installed in the station is made to vary as a function of the number of output ports on the secondary side or the connectors attached to each of them and their individual rated power. In order to lessen the deteriorating impact on the distribution system, some indices are included in the objective function.

It is assumed that an area equal to $25m^2$ is required for each connector. It is assumed that the present work is a pilot project for the city and hence the land is assumed to be on rent for a period of 5 years. The operating and maintenance cost of the project is hence neglected for this study. The initial development cost (IC_i) of the i^{th} station is [17]:

$$IC_i = C_{in} + 25 \times C_{land} \times N_i + C_p \times C_{dev} \times (N_i - 1) \quad (1)$$

where C_p is the rated power of each connector, C_{dev} is the cost associated with development of connector, C_{land} is the rental cost of land for 5 years, and N_i is the number of connectors in i^{th} station.

The cost of land can be varied in the present method depending on the location in the city. The development cost of each connector, C_{dev} will decrease in the future as the technology progresses, since it forms a major part of the total cost of the project. The initial investment takes into account the cost of planning the infrastructure, raw materials used, labour cost etc.

The total capacity of the i^{th} charging station S_{cap}^i is given by:

$$S_{cap}^i = C_p \times N_i \quad (2)$$

The cost C_p includes the power electronics components and other utilities which are necessary for a charging station. As mentioned in [22], the cost of such equipment's varies slightly in different regions and countries. Figure 1 presents a representative 30 bus distribution system of a substation in Allahabad city.

In order to ensure that the power quality and reliability of the nearby distribution line, three stability indices have been defined [23].

i) *Real Power Loss Reduction Index (PLRI)*: Connection of EVCS to the grid increases the power losses. In order to minimize the losses at each point of connection of charging station, *PLRI* has been incorporated in the mathematical objective function, defined as:

$$PLRI_i = \frac{P_{L(base)} - P_{L(CS_i)}}{P_{L(base)}} \quad (3)$$

where, $P_{L(base)}$ equals the active power losses of system before connection of CS (calculated using load flow equations), $P_{L(CS_i)}$ equals active power loss of i^{th} bus after connection of CS.

ii) *Reactive Power Loss Reduction Index (QLRI)*: Voltage stability of the grid is retained when reactive power losses in the system are below a threshold value. For achieving this objective, *QLRI* has been included in the objective function, defined as:

$$QLRI_i = \frac{Q_{L(base)} - Q_{L(CS_i)}}{Q_{L(base)}} \quad (4)$$

where, $Q_{L(base)}$ equals the reactive power losses before connection of CS (calculated using load flow equations). $Q_{L(CS_i)}$ equals the reactive power losses after connection of CS.

iii) *Voltage Profile Improvement Index (VPPI)*: Voltage at each base (node) should remain within acceptable limits as large deviations may cause unnecessary increase in the cost of voltage control equipments or change in grid infrastructure. This index imposes a penalty factor on any location-size pair of CS which causes substantial voltage deviations from the original base voltage value.

$$VPPI_i = \frac{1}{\alpha + \max_{i=2}^n (|1 - V_{(CS_i)}|)} \quad (5)$$

where, $V_{(CS_i)}$ is voltage at i^{th} bus after connection of CS. α is a scalar value.

2.1 Mathematical Model

The objective function to minimize the setup cost and sustenance of grid power quality parameters such as voltage profile and power losses is incorporated in the function as:

$$F_i = \frac{1}{w_1 VPII_i + w_2 PLRI_i + w_3 QLRI_i} \quad (i \in NC) \quad (6)$$

where, w_1, w_2 and w_3 are the weights assigned to each index. It should be noted that:

$$|w_1| + |w_2| + |w_3| = 1 \quad (7)$$

The final objective function comes out to be

$$F_{min} = \sum_{i \in NC} IC_i X_i + \sum_{i \in NC} F_i \quad (8)$$

where, X_i represents the binary variable (0 or 1) which decides whether the particular node is suitable for station placement. NC is the cardinal number of prospective charging stations set to be located in the city.

2.2 Constraints

The above formulated objective function is minimized in agreement with various operational constraints which ensure that the planning of the charging station infrastructure doesn't compromise the electrical grid of Allahabad. Due to nonlinear load characteristics of charging devices, it gives higher order harmonics in the utility. Compensation using filters/reactive banks increases cost. The following inequality constraints are imposed in developing the framework for optimal siting and sizing of EVCS.

- 1) The voltage levels at the buses should remain within set standard limits so that the grid stability is retained.

$$V_i^{\min} \leq V_i \leq V_i^{\max} \quad (i = 1, 2, \dots, N) \quad (9)$$

where, $V_i^{\min}$ and $V_i^{\max}$ represent the value of minimum and maximum voltage level.

- 2) A limit is imposed on the minimum and maximum levels of the real and reactive power levels on the i^{th} bus.

$$P_{gi}^{\min} \leq P_{gi} \leq P_{gi}^{\max} \quad (i = 1, 2, \dots, N_{gen}) \quad (10)$$

$$Q_{gi}^{\min} \leq Q_{gi} \leq Q_{gi}^{\max} \quad (i = 1, 2, \dots, N_{gen}) \quad (11)$$

where, $P_{gi}^{\min}$ and $P_{gi}^{\max}$ equals the maximum and minimum limits on real power generation. $Q_{gi}^{\min}$ and $Q_{gi}^{\max}$ equals the maximum and minimum limits on reactive power generation.

- 3) There should be no mismatch in the load and generation power, i.e. load should be properly regulated during charging of electric vehicles.

$$P_{gni} - P_{dni} - V_{ni} \sum_{j=1}^N V_{nj} Y_{nj} \cos(\delta_{ni} - \delta_{nj} - \theta_{nj}) = 0 \quad (12)$$

$$Q_{gni} - Q_{dni} - V_{ni} \sum_{j=1}^N V_{nj} Y_{nj} \sin(\delta_{ni} - \delta_{nj} - \theta_{nj}) = 0$$

(13) where, $n_i = 1, 2, \dots, n_n$

- 4) It is also necessary that at least one charging station should be placed at each **optimally selected** node i.e.:

$$\sum_{i \in NC} X_i > 0 \quad (14)$$

- 5) The planning infrastructure should ensure that each charging station has at least one connector installed:

$$N_i \geq X_i \quad (i = 1, 2, \dots, NC) \quad (15)$$

- 6) Another important consideration is the maximum number of connectors any station can install. The number of e-rickshaws nearby candidate CS, population density and nature of locality (residential, industrial etc.) presently in Allahabad at the time when this work was in progress is used to estimate the maximum limit on the connector in each station. The ability of distribution system to supply required power is another basis for this constraint:

$$N_i^{\min} \leq N_i \leq N_i^{\max} \quad (i = 1, 2, \dots, NC) \quad (16)$$

3. OBJECTIVE FUNCTION OPTIMIZER

It is evident from the proposed objective function in (8) and the constraints that follow in (9-16) that the present problem is constrained Mixed Integer Non-Linear Problem (MINLP). Using classical mathematical programming techniques to solve this type of problem won't be feasible, hence evolutionary algorithms are considered for the present problem. Over the past few years, Genetic algorithm (GA) and Particle Swarm Optimization (PSO) have been implemented in [7], [17], [20] and [21]. Figure 2 depicts a flowchart of the GAIPSO algorithm.

3.1 Algorithm

Step 1: Input bus and line data for given distribution system.

Step 2: Calculate system's base power loss through load flow studies using Newton-Raphson method.

Step 3: Input IPSO and GA parameters

Step 4: Initiate by setting bus count of candidate $i=1$

Step 5: Set loop construct

while $i \leq I$

- i. Randomly allot N chromosomes (possible solution) with values representing size of the charging station (CS) (specifically, number of charger ports)

$$P_{CS}^{\min} \leq P_{CSj} \leq P_{CS}^{\max} \quad (17)$$

and

$$Q_{CS}^{\min} \leq Q_{CSj} \leq Q_{CS}^{\max} \quad (18)$$

where $j=1, 2, \dots, N$

- ii. For GA, set count of iteration $k=1$
- iii. while $k \leq k_{max}$
 - a) Evaluate fitness value of each chromosome employing the multi-objective function
 - b) Employ method of roulette wheel selection to choose two chromosomes (X_{pa1} and X_{pa2})
 - c) Implement crossover and mutation in accordance with probabilities P_{mut} and P_{cross}
 - d) Repeat step (b) and (c) to establish new population and simultaneously store newly created children in the already existing population until the new population is completely generated again.
 - e) Substitute old population with the one created in (d).
 - f) Update iteration value $k=k+1$
- iv. Pass the present version of the partially optimized chromosomes to IPSO.
- v. Treat the chromosomes optimized by GA as the initial particles for IPSO.
- vi. Evaluate each individual particle's fitness value utilizing the multi-objective function. Each particle's fitness value equals p_{best} and the minimum amongst the computed set of fitness values is designated as q_{best}
- vii. Choose the iteration count for IPSO as $iter=1$

while $iter \leq iter_{max}$

- a) Update the particle's velocity as:

$$V_{id}^{k+1} = wV_{id}^k + c_1r(P_{best_{id}} - S_{id}^k) + c_2r(G_{best_{id}} - S_{id}^k) \quad (19)$$

where, weight w is varied linearly by

$$w_{iter} = w_{max} - \frac{(w_{max} - w_{min})}{iter_{max}} \cdot iter \quad (20)$$

b) Update each particle's position

$$S_{id}^{iter+1} = S_{id}^{iter} + V_{id}^{iter+1} \quad (21)$$

c) Choose two chromosomes through greedy selection method, namely S_{p1} and S_{p2} .

d) Carry out crossover and mutation operation using probabilities P_{cross} and P_{mut} .

e) Repeat steps (c) and (d) to form a new population while simultaneously acquiring newly formed children until the new population is complete.

f) Evaluate fitness of individual new particle and update position, p_{best} and velocity, q_{best} as shown:

$$p_{best(j)}^{iter+1} = \begin{cases} S_{(j)}^{iter+1} & \text{if } MOF_j^{iter+1} < MOF_j^{iter} \\ p_{best(j)}^{iter} & \text{if } MOF_j^{iter+1} \geq MOF_j^{iter} \end{cases} \quad (22)$$

$$q_{best(j)}^{iter+1} = \begin{cases} p_{best(j)}^{iter+1} & \text{if } MOF_j^{iter+1} < MOF_j^{iter} \\ q_{best}^{iter} & \text{if } MOF_j^{iter+1} \geq MOF_j^{iter} \end{cases} \quad (23)$$

g) Update $iter=iter+1$

viii. Latest q_{best} of the particle is the optimal solution of the problem.

ix. Using q_{best} compute bus voltages and system power loss.

x. Update candidate bus $i=i+1$.

Step 6: Compare the resultant fitness of each candidate bus, q_{best} and find out the particle with minimum fitness value.

4.CASE STUDY

The proposed algorithm is tested in MATLAB 2012b software installed on computer with the processor of 64-bit Intel Xeon CPU @ 3 GHz. The performance of the proposed GAIPSO method is tested on the 220kV/132kV transformer feeds Allahabad city distribution substation was shown in Figure 2. The objective function parameters used for the case study is shown in the Table 1.

The voltage and load profile of the bus system considered for the case study has been listed in Table 2. This system has 21 load buses in which 14 potential CS have been selected on the basis of a detailed survey considering electricity availability, space convenience and nature of nearby localities i.e., residential, commercial etc., through which the intensity of transportation required was judged.

A base value of 100 MVA and 30 kV is assumed for ease of calculations. As information obtained from local EV charger traders in the city, the rated power of a CS connector is taken to be 4.8 kW. No official record for the number of electric vehicles registered in Allahabad and after visiting the shops in the city,

approximately 1500 vehicles were considered. Figure 3 shows the position of optimally selected charging stations on the distribution bus system.

Figure 4 depicts the actual position of the selected charging stations on a map of Allahabad.

4.1 Parameter Selection and its Effect

The parameters for GAIPSO method are given in Table 3. To find optimal location and sizing of CS, GA initially will do space exploration and computes a sub-optimal solution set to PSO, thereby enhancing the operational functionality of the conventional PSO algorithm. The proposed GAIPSO method further tunes the solution set and produces an optimal solution unlike conventional optimization. The limit on the particle velocity, v_{min} and v_{max} guarantees that the proposed GAIPSO does not get entrapped in local minima. Learning coefficients are made equal to 2. The inertia weight is also constrained with minimum and maximum limit as 0.2 and 0.9 which achieves a balance between exploration and exploitation.

4.2 Evaluation of optimal weights for study

To demonstrate the effectiveness of proposed method seven test cases were performed for different values of w_1 , w_2 and w_3 . Table 4 represents optimal sizing of CS chargers, system active power and reactive power losses in the Allahabad city distribution system. The table outlines the results for different values of w_i ($i \in [1, 3]$) to be employed in objective function. Case I with $w_1=0.2$ $w_2=0.6$ and $w_3=0.2$ satisfies the city's charging requirements with acceptable losses in the distribution system. For case 1, the power loss is comparatively higher than other cases because more number of chargers is placed in the distribution system.

Table 5 outlines the optimal allocation plan suggested by proposed GAIPSO algorithm. It is self-explanatory that more charging stations are feasible to be installed in the distribution system near sub-feeder 5 and 6. From Table 1, the area around these sub-feeders can be classified as residential area as the loading on the feeder is less as compared to other sub-feeders. Sub-feeder 1 and 2 constitutes the heavy industrial region and Sub-feeder 3 and 4 covers the light industrial region. The area requirement near the residential areas is generally scarcer and hence the node 4.3 (light industrial region) has the maximum number of CS. The optimum number of chargers which should be installed at each station is mentioned. The present rate of charging, each vehicle takes 6 hours to get fully charged. At this rate, the total number of vehicles that can be charged in 24 hours is $N_{EV's} = 412 \times (24/6) = 1648$, which meets the actual demand at present in the city. The EV transportation mode would get popular in future, which cause vehicle charger manufacturers could switch over towards the standard Level II (AC charging) or Level III (DC charging) in order to reduce the charging time per vehicle and facilitate charging of more electric vehicles.

4.3 Validation of the Proposed Method

To validate the proposed algorithm, its performance is compared with conventional optimization algorithm GA and PSO. The voltage profile of the system is shown in Figure 5. Optimal allocation plan with due consideration to voltage constraints achieves lesser peaking in the voltage levels, with smaller voltage deviation between nearby buses. It is evident from the figure that the voltage profile is smoothened and a logical tradeoff between network losses and voltage profile is achieved.

Table 6 gives a comparative analysis of GA, PSO and GAIPSO for the given study. It is evident that the proposed algorithm demonstrates superior performance with lesser power losses. The number of CS allocated through GA and PSO in CS doesn't satisfy the demand of the city's consumers.

Figure 6 further proves that the present approach is capable of establishing a charging infrastructure with least possible harm to the existing grid network. The voltage level remains within the maximum and minimum specified limits of 1.1 and 0.95 p.u., respectively. Reduction in the peak value reduces stress on transformers and reduces the need for costly protective devices. This validates that the present plan is effective enough to be implemented in a real world environment.

The parameters of GA and PSO are same as the one used in this paper separately. Figure 7 shows the convergence rate versus iteration plot for all the three methods. It is evident that the proposed method finds the optimal solution within 20 iterations (exactly 14) whereas PSO and GA converged after 34 and 30 iterations respectively producing sub-optimal solutions.

4.4 Result Discussion

The proposed method would be helpful in developing the electric vehicle infrastructure with minimum costs associated with planning, development and maintenance. The stress on the existing power network due to inclusion of the charging stations is minimal with the proposed method of solving the optimization problem with GAIPSO. Another important point is the reachability of these stations to urban consumers as the charging stations are optimally located in areas with high residential areas and inter-city electric transportation services. With low levels of power losses, it is a viable solution to bolster the electric vehicle scenario in India.

5. CONCLUSION

The paper presented a novel optimal placement strategy of electric vehicles charging station. A hybrid algorithm which inherited the best features of both GA and PSO was used considering the initial investment cost and distribution grid power quality as another parameter in the objective function. Candidate stations were chosen after a detailed survey using which a case study was performed to find the

optimal location for setting up such stations. Voltage profile was found to suffice the constraints and showed improvement in the lowest p.u. voltage value. A better performance in terms of quality of solution was obtained when compared to both GA and PSO with lesser number of iterations. The number of charging stations met the required criteria while imposing minimum stress on the distribution system. Future work will deal with vehicle to grid (V2G) operation of charger and long term smart planning of fast charging infrastructure for Allahabad city.

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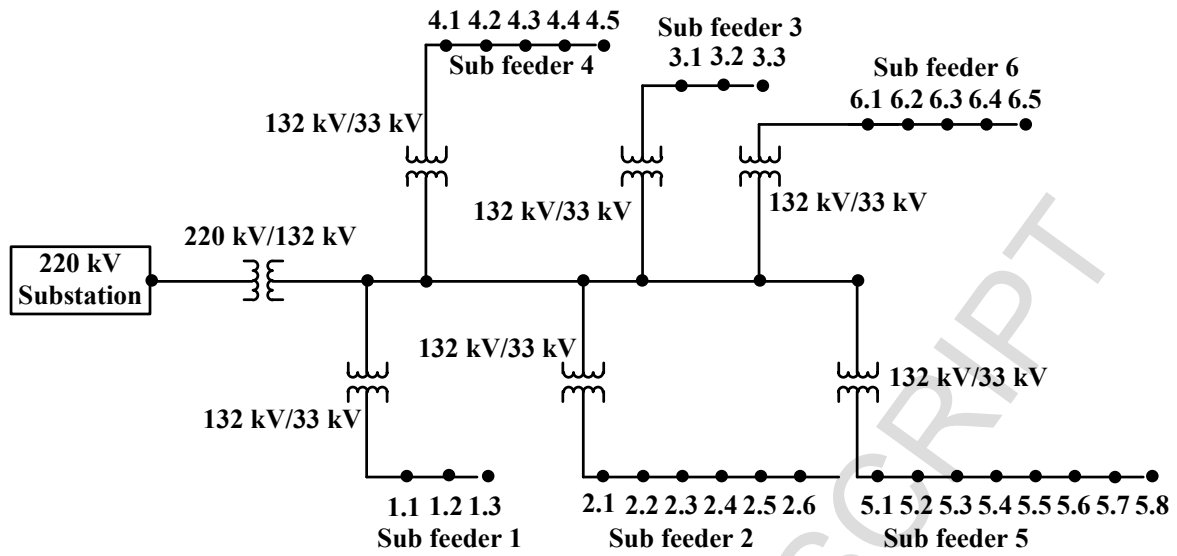


Figure 1 Radial Distribution system of a substation of Allahabad city

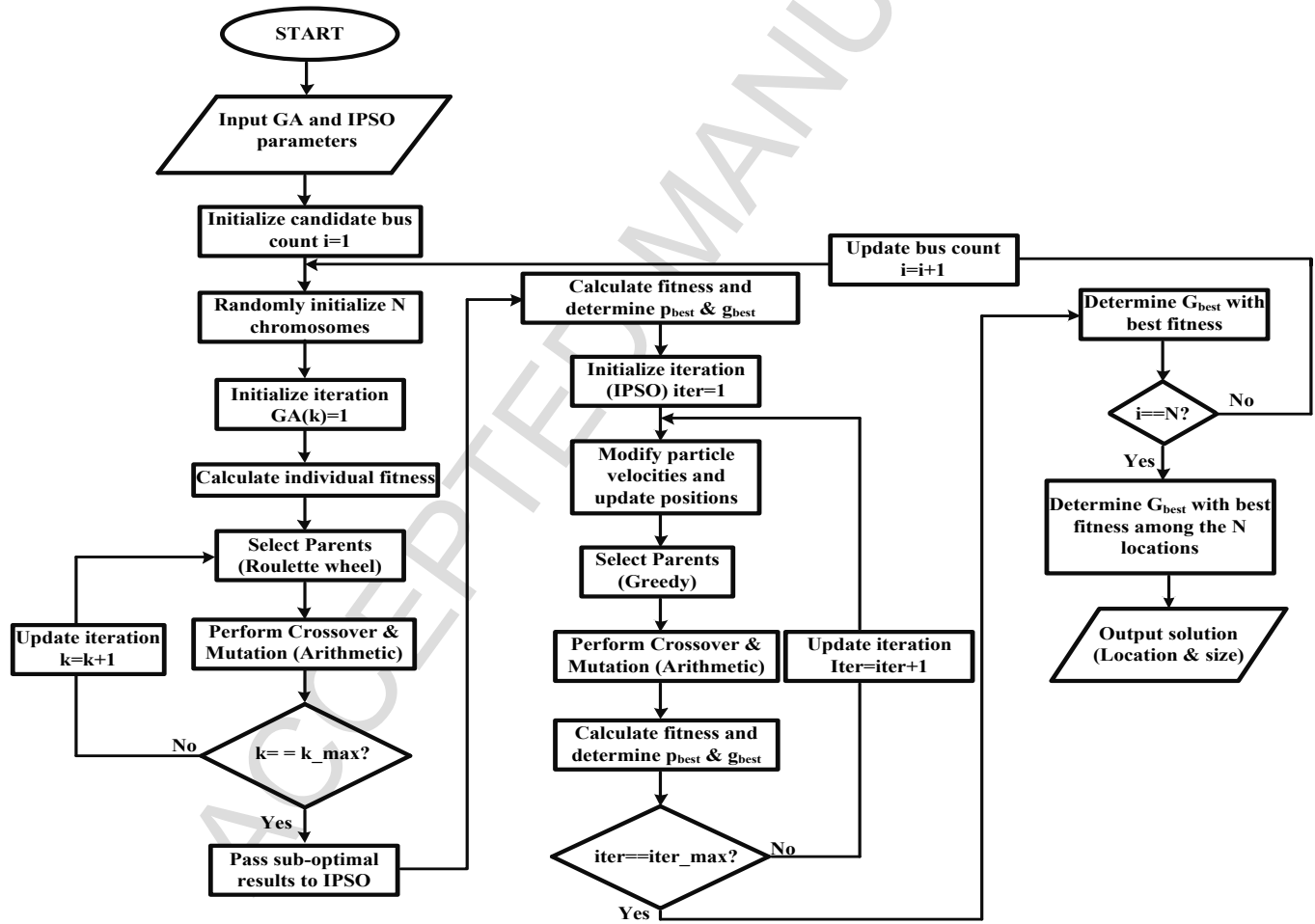


Figure 2. Flow chart of proposed algorithm

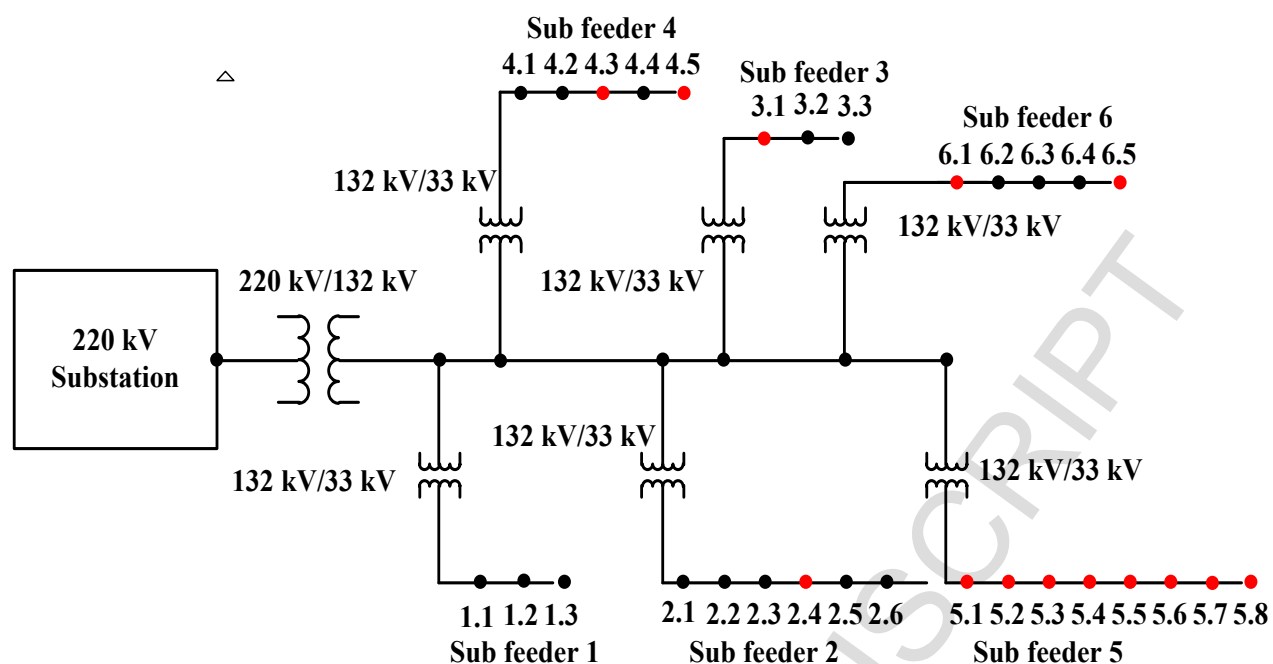


Figure 3 Location of optimal charging station (Red) on the distribution system



Figure 4. Google Map showing the location of optimal charging station (Blue triangles)

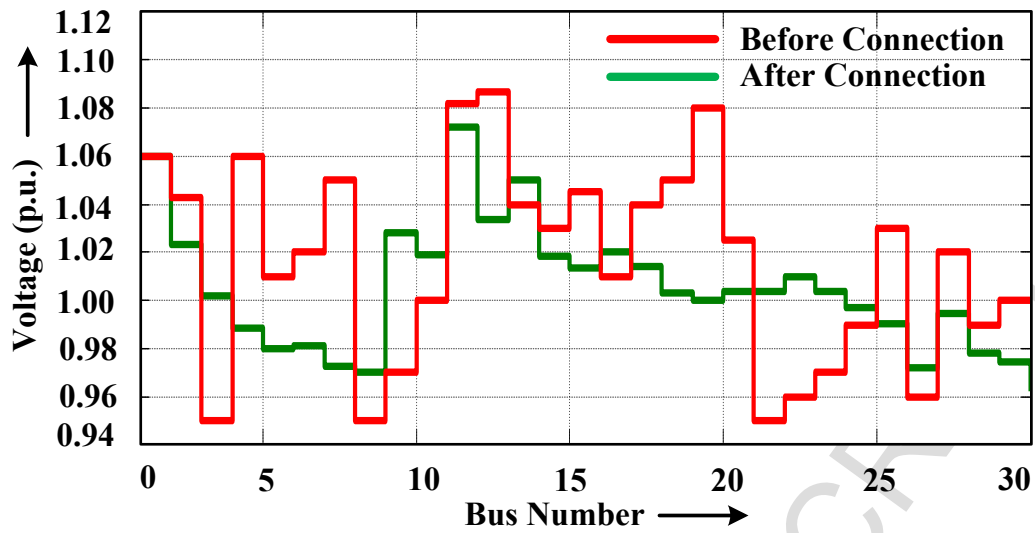


Figure 5. Voltage profile of the Allahabad bus system.

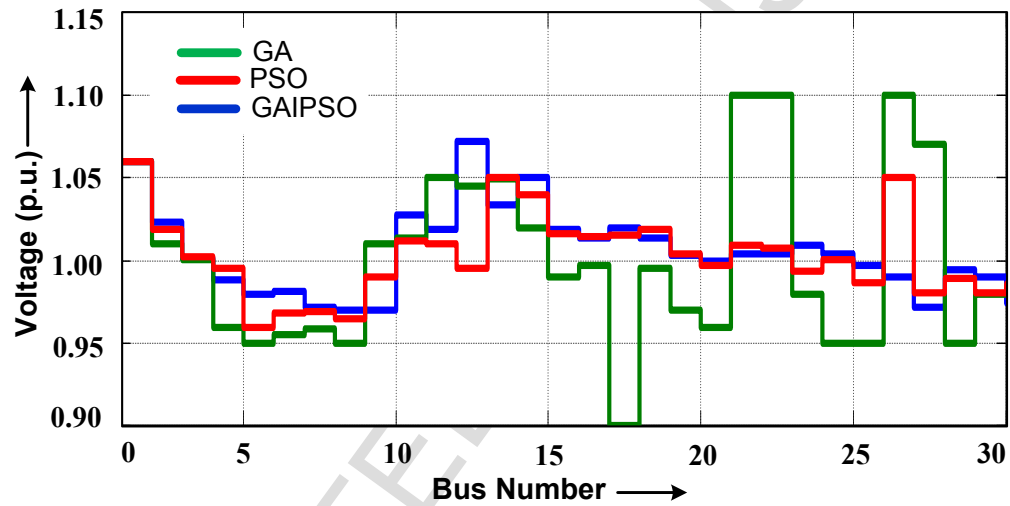


Figure 6. Voltage profile of Real time Allahabad bus system for GA, PSO and GAIPSO.

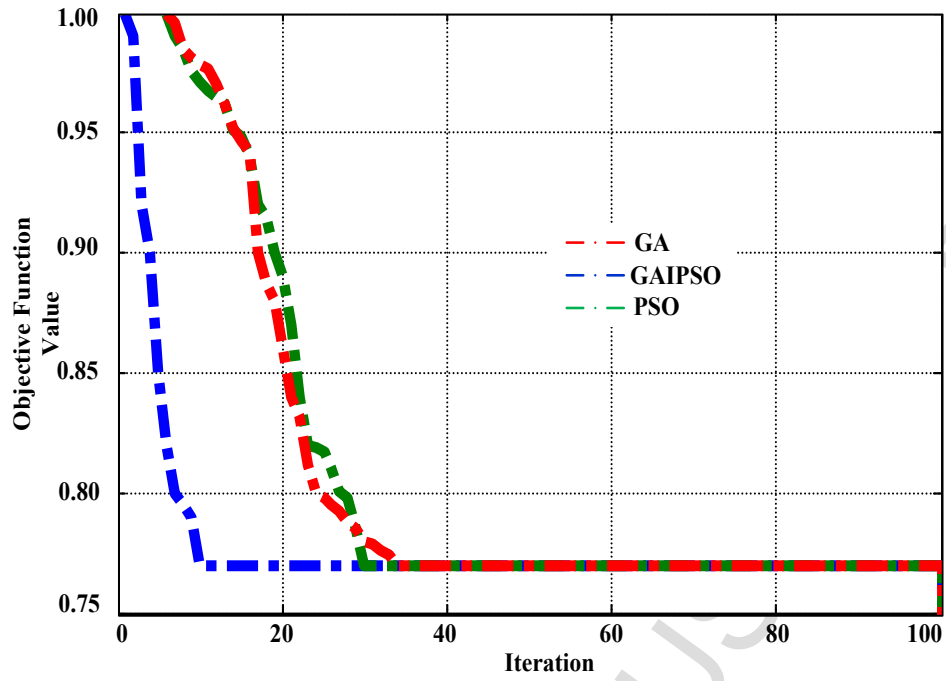


Figure 7. Convergence characteristics of GA, PSO and GAIPSO.

TABLE 1
OBJECTIVE FUNCTION PARAMETERS

Parameter	Value	Unit
C_{in}	70,000 [25]	\$
C_{dev}	208.33 [25]	\$/kW
C_p	4.8	kW
NC	14	

TABLE 2
VOLTAGE AND LOAD PROFILE OF 30-BUS DISTRIBUTION SYSTEM
1-SLACK BUS 2-GENERATOR BUS 3-LOAD BUS

Bus	Type	$V_{sp}(p.u.)$	$PL_i(MW)$	$QL_i(MVar)$	Region Type
1.1	1	1.06	0	0	Heavy Industrial Region
1.2	2	1.043	21.7	12.7	
1.3	3	0.95	2.4	1.2	
2.1	3	1.06	7.6	1.6	
2.2	2	1.01	94.2	19.0	
2.3	3	1.02	0	0	
2.4	3	1.05	22.8	10.9	
2.5	2	0.95	30	30	
2.6	3	0.97	0	0	Light Industrial Region
3.1	3	1.0	5.8	2.0	
3.2	2	1.082	0	0	
3.3	3	1.087	11.2	7.5	

4.1	2	1.04	0	0	
4.2	3	1.03	6.2	1.6	
4.3	3	1.045	8.2	2.5	
4.4	3	1.01	3.5	1.8	
4.5	3	1.04	9.0	5.8	
5.1	3	1.05	3.2	0.9	Residential Region
5.2	3	1.08	9.5	3.4	
5.3	3	1.025	2.2	0.7	
5.4	3	0.95	17.5	11.2	
5.5	3	0.96	0	0	
5.6	3	0.97	3.2	1.6	
5.7	3	0.99	8.7	6.7	
5.8	3	1.03	0	0	
6.1	3	0.96	3.5	2.3	
6.2	3	1.02	0	0	
6.3	3	0.99	0	0	
6.4	3	1.0	2.4	0.9	
6.5	3	1.0	10.6	1.9	

TABLE 3
GAIPSO PARAMETERS

Parameter	Value
No. of candidate charging station (N_p)	14
No. of dimensions (N_d)	2
No. of trials (N_t)	100
Minimum velocity (v_{min})	0
Maximum velocity (v_{max})	0.015
Maximum inertia weight (w_{max})	0.9
Minimum inertia weight (w_{min})	0.2
Learning factor ($c_1=c_2$)	2

TABLE 4
COMPARISON ANALYSIS OF WEIGHTS WITH LOSSES AND QUANTITY OF CHARGERS

Case	w_1	w_2	w_3	Active Power Loss (MW)	Reactive Power Loss (MVar)	No. of EV chargers
1	0.2	0.6	0.2	18.532	73.989	412
2	0.3	0.6	0.1	18.296	72.967	57
3	0.1	0.6	0.3	18.325	73.092	112
4	0.4	0.5	0.1	18.438	73.581	259
5	0.1	0.5	0.4	18.307	73.014	85
6	0.2	0.5	0.3	18.460	73.676	276
7	0.3	0.5	0.2	18.414	73.476	225

TABLE 5

OPTIMAL ALLOCATION PLAN FOR EVCS

Location	No. of chargers	Area(m ²)	Cost (\$) $\times 10^5$
2.4	29	725	1.0042
3.1	29	725	0.9921
4.3	41	1025	1.1512
4.5	31	775	1.0646
5.1	28	700	0.9758
5.2	28	700	0.9846
5.3	28	700	1.0003
5.4	28	700	0.9700
5.5	28	700	0.9787
5.6	28	700	0.9860
5.7	28	700	0.9962
5.8	28	700	1.0108
6.1	29	725	1.0011
6.5	29	725	1.0283

TABLE 6

PERFORMANCE OF EVOLUTIONARY ALGORITHMS

Algorithm	Active Power Loss (MW)	Reactive Power Loss (MVar)	Computational Burden(sec.)	Optimal Size of CS (Number of Charging Ports)
GAIPSO	18.532	73.989	11128.7	412
GA	18.992	74.788	9678	200
PSO	18.724	74.189	10117	333